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LAB EXERCISE 15- CREATING CONFIGMAPS IN KUBERNETES

IN THIS LAB EXERCISE, YOU WILL LEARN HOW TO CREATE CONFIGMAPS IN KUBERNETES USING DIFFERENT METHODS: FROM LITERAL VALUES, FROM A FILE, FROM AN ENVIRONMENT FILE, AND FROM MULTIPLE FILES. CONFIGMAPS ALLOW YOU TO DECOUPLE CONFIGURATION ARTIFACTS FROM IMAGE CONTENT TO KEEP CONTAINERIZED APPLICATIONS PORTABLE.

STEP 1: SET UP KUBERNETES CLUSTER

ENSURE YOU HAVE ACCESS TO A KUBERNETES CLUSTER. YOU CAN USE A LOCAL SETUP WITH MINIKUBE, KIND, OR USE A CLOUD-BASED KUBERNETES SERVICE.

STEP 2: CREATE A CONFIGMAP FROM LITERAL VALUES

CREATE A CONFIGMAP NAMED EXAMPLE-CONFIGMAP WITH LITERAL VALUES:

```
KUBECTL CREATE CONFIGMAP EXAMPLE-CONFIGMAP --FROM-LITERAL=KEY1=VALUE1 --  
FROM-LITERAL=KEY2=VALUE2  
--from-literal=key2=value2  
configmap/example-configmap created
```

VERIFY THE CONFIGMAP:

```
KUBECTL GET CONFIGMAP EXAMPLE-CONFIGMAP -O YAML  
apiVersion: v1  
data:  
  key1: value1  
  key2: value2  
kind: ConfigMap  
metadata:  
  creationTimestamp: "2026-03-02T04:46:18Z"  
  name: example-configmap  
  namespace: default  
  resourceVersion: "5960"  
  uid: 388d5351-182d-4aed-8039-397f7269a035
```

YOU SHOULD SEE THE EXAMPLE-CONFIGMAP WITH THE KEYS AND VALUES SPECIFIED.

STEP 3: CREATE A CONFIGMAP FROM A FILE

CREATE A FILE NAMED **CONFIG.TXT** WITH THE FOLLOWING CONTENT:

```
KEY1=VALUE1
KEY2=VALUE2

key1=value1
key2=value2
```

CREATE A CONFIGMAP NAMED FILE-CONFIGMAP FROM THE FILE:

```
KUBECTL CREATE CONFIGMAP FILE-CONFIGMAP --FROM-FILE=CONFIG.TXT
```

VERIFY THE CONFIGMAP:

```
KUBECTL GET CONFIGMAP FILE-CONFIGMAP -O YAML
```

```
apiVersion: v1
data:
  config.txt: |
    key1=value1
    key2=value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T04:47:07Z"
  name: file-configmap
  namespace: default
  resourceVersion: "6000"
  uid: 57f228cd-142b-449a-845d-192fa3d0bcc5
```

YOU SHOULD SEE THE FILE-CONFIGMAP WITH THE CONTENT OF CONFIG.TXT.

STEP 4: CREATE A CONFIGMAP FROM AN ENVIRONMENT FILE

CREATE AN ENVIRONMENT FILE NAMED **ENV-CONFIG.ENV** WITH THE FOLLOWING CONTENT:

```
ENV_VAR1=VALUE1
ENV_VAR2=VALUE2

ENV_VAR1=value1
ENV_VAR2=value2
```

CREATE A CONFIGMAP NAMED ENV-CONFIGMAP FROM THE ENVIRONMENT FILE:

```
KUBECTL CREATE CONFIGMAP ENV-CONFIGMAP --FROM-ENV-FILE=ENV-CONFIG.ENV
```

VERIFY THE CONFIGMAP:

```
KUBECTL GET CONFIGMAP ENV-CONFIGMAP -O YAML
```

```
apiVersion: v1
data:
  ENV_VAR1: value1
  ENV_VAR2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T04:48:34Z"
  name: env-configmap
  namespace: default
  resourceVersion: "6066"
  uid: abf47a1a-fbe6-44d9-81a6-c1d893a69a35
```

YOU SHOULD SEE THE ENV-CONFIGMAP WITH THE ENVIRONMENT VARIABLES SPECIFIED IN THE FILE.

STEP 5: CREATE A CONFIGMAP FROM MULTIPLE FILES

CREATE MULTIPLE FILES:

CONFIG1.TXT WITH CONTENT:

```
KEY1=VALUE1
```

CONFIG2.TXT WITH CONTENT:

```
KEY2=VALUE2
```

```
key1=value1
key2=value2
```

CREATE A CONFIGMAP NAMED MULTI-FILE-CONFIGMAP FROM MULTIPLE FILES:

```
KUBECTL CREATE CONFIGMAP MULTI-FILE-CONFIGMAP --FROM-FILE=CONFIG1.TXT --
FROM-FILE=CONFIG2.TXT
```

VERIFY THE CONFIGMAP:

```
KUBECTL GET CONFIGMAP MULTI-FILE-CONFIGMAP -O YAML
```

```
apiVersion: v1
data:
  config1.txt: |
    key1=value1
  config2.txt: |
    key2=value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T04:50:06Z"
  name: multi-file-configmap
  namespace: default
  resourceVersion: "6140"
  uid: 550370d2-1253-474a-86f5-0a9571dc692b
```

YOU SHOULD SEE THE MULTI-FILE-CONFIGMAP WITH THE CONTENT OF BOTH CONFIG1.TXT AND CONFIG2.TXT.

STEP 6: USE CONFIGMAPS IN A POD

CREATE A FILE NAMED **POD-CONFIGMAP.YAML** WITH THE FOLLOWING CONTENT:

```
APIVERSION: v1
KIND: POD
METADATA:
  NAME: CONFIGMAP-POD
SPEC:
  CONTAINERS:
  - NAME: NGINX
    IMAGE: NGINX
    ENVFROM:
    - CONFIGMAPREF:
      NAME: ENV-CONFIGMAP
```

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
      name: env-configmap
```

IN THIS MANIFEST:

- THE POD READS ENVIRONMENT VARIABLES FROM THE ENV-CONFIGMAP.
- THE POD MOUNTS THE FILE-CONFIGMAP AT /ETC/CONFIG.

APPLY THE MANIFEST TO CREATE THE POD:

```
KUBECTL APPLY -F POD-CONFIGMAP.YAML
```

CHECK THE STATUS OF THE POD:

KUBECTL GET PODS

VIEW THE POD LOGS TO SEE THE ENVIRONMENT VARIABLES AND FILE CONTENT:

KUBECTL EXEC -IT CONFIGMAP-POD – PRINTENV

```
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
ENV_VAR1=value1
ENV_VAR2=value2
KUBERNETES_PORT=tcp://10.96.0.1:443
NODEPORT_SERVICE_PORT_80_TCP_PROTO=tcp
NODEPORT_SERVICE_SERVICE_HOST=10.101.115.216
NODEPORT_SERVICE_SERVICE_PORT=80
NODEPORT_SERVICE_PORT=tcp://10.101.115.216:80
NODEPORT_SERVICE_PORT_80_TCP=tcp://10.101.115.216:80
NODEPORT_SERVICE_PORT_80_TCP_ADDR=10.101.115.216
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NODEPORT_SERVICE_PORT_80_TCP_PORT=80
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
NJS_RELEASE=1-trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1-trixie
DYNPKG_RELEASE=1-trixie
HOME=/root
```

YOU SHOULD SEE THE ENVIRONMENT VARIABLES AND THE CONTENT OF THE FILE-CONFIGMAP.

STEP 7: CLEAN UP

AFTER COMPLETING THE EXERCISE, CLEAN UP THE RESOURCES CREATED:

```
KUBECTL DELETE POD CONFIGMAP-POD
```

```
KUBECTL DELETE CONFIGMAP EXAMPLE-CONFIGMAP
```

```
KUBECTL DELETE CONFIGMAP FILE-CONFIGMAP
```

```
KUBECTL DELETE CONFIGMAP ENV-CONFIGMAP
```

```
KUBECTL DELETE CONFIGMAP MULTI-FILE-CONFIGMAP
```

```
RM CONFIG.TXT ENV-CONFIG.ENV CONFIG1.TXT CONFIG2.TXT
```